

META AD PERFORMANCE ANALYSIS DASHBOARD

Project Documentation

Platform: Microsoft Power BI

Advertising Analytics | Campaign Intelligence | Audience Insights

1. Project Overview

The Meta Ad Performance Analysis Dashboard is an interactive Business Intelligence project built using Microsoft Power BI. It analyzes and monitors advertising performance across Meta platforms — primarily Facebook and Instagram — transforming raw event data into clear, actionable marketing insights through dynamic visualizations.

1.1 What This Dashboard Does

This dashboard enables users to:

- Analyze advertising performance across Meta platforms
- Compare Facebook and Instagram ad performance side by side
- Dynamically switch between multiple event metrics using a single slicer
- Track engagement and conversion KPIs in real time
- Analyze audience demographics (age, gender, interests, location)
- Monitor campaign performance trends over time
- Generate actionable, data-driven marketing insights

1.2 Technical Skills Demonstrated

The project showcases end-to-end BI development capabilities, including:

- ETL Process — Extract, Transform, Load pipeline
- Data Cleaning & Transformation using Power Query
- Data Modeling — Hybrid schema structure
- Dynamic Measure Parameters for flexible KPI switching
- DAX Calculations — Custom measures and calculated columns
- Time Intelligence — Weekly and monthly trend analysis
- Interactive Dashboard Design with tooltips and platform toggles
- KPI Analytics — Rate measures such as CTR, CR, ER, PR

2. Project Objective

The primary objective of this dashboard is to help marketers and businesses measure the effectiveness of their Meta advertising campaigns and optimize their strategy based on data.

The dashboard provides insights across the following areas:

Insight Area	Key Metrics Covered
Performance Metrics	Impressions, Clicks, Shares, Comments, Purchases, Engagements
Conversion Metrics	CTR, Engagement Rate, Conversion Rate, Purchase Rate
Audience Demographics	Age groups, Gender, Country, User Interests
Ad Type Performance	Platform-level breakdown by ad format

3. Tools & Technologies

Tool / Technology	Purpose
Microsoft Power BI	Dashboard development and visualization
Power Query	ETL pipeline and data transformation
DAX (Data Analysis Expressions)	Custom measures, calculated columns, and KPIs
Excel / CSV	Source data files
Data Modeling	Relationship management between tables
Dynamic Parameters	Dynamic KPI and visual switching via slicers
Time Intelligence	Weekly and monthly trend analysis

4. Dataset Description

The project is built on a multi-table data model consisting of one Fact Table and four Dimension Tables. Each table serves a distinct analytical purpose.

4.1 Fact Table — ad_events

This is the central table that stores all advertisement event data. Every user interaction (impression, click, purchase, etc.) is recorded as a row in this table.

Column Name	Description
event_id	Unique identifier for each event record
ad_id	Links the event to a specific advertisement
user_id	Links the event to a specific user
timestamp	Date and time when the event occurred
day_of_week	Name of the day (e.g., Monday, Tuesday)
time_of_day	Time category (e.g., Morning, Evening)
event_type	Type of interaction (Impression, Click, Purchase, etc.)

4.2 Dimension Tables

4.2.1 ads — Advertisement Details

Column	Description
ad_id	Unique advertisement identifier (Primary Key)
campaign_id	Links the ad to its parent campaign
ad_platform	Platform where the ad ran (Facebook / Instagram)
ad_type	Format of the advertisement (e.g., video, image)
target_gender	Gender audience targeted by the ad
target_age_group	Age group audience targeted by the ad
target_interests	Interest-based audience targeting categories

4.2.2 campaigns — Campaign Information

Column	Description
campaign_id	Unique campaign identifier (Primary Key)
name	Descriptive name of the campaign
start_date	Campaign launch date
end_date	Campaign end date
duration_days	Total number of days the campaign ran
total_budget	Total budget allocated to the campaign

4.2.3 users — User Demographics

Column	Description
user_id	Unique user identifier (Primary Key)
user_gender	Gender of the user
user_age	Exact age of the user
age_group	Age category (e.g., 18-24, 25-34)
country	Country where the user is located
location	Specific location or city of the user
interests	User's listed interest categories

4.2.4 Calendar Table

A dedicated Calendar Table was created using the event_date column from the ad_events fact table. This enables all Time Intelligence functions within Power BI, including weekly trends, monthly analysis, and date-based filtering.

5. ETL Process

The project follows a complete ETL (Extract, Transform, Load) pipeline to prepare raw data for analysis.

5.1 Extract

Raw advertising datasets were imported directly into Power BI from Excel/CSV source files.

5.2 Transform

Data cleaning and transformation were performed using Power Query. The following activities were carried out:

- Corrected incorrect data types across all columns
- Removed inconsistent or invalid values
- Standardized date and time formats for uniformity
- Verified key relationship columns (foreign keys) for data integrity
- Cleaned missing, null, or malformed records

5.3 Load

After transformation, all cleaned datasets were loaded into the Power BI data model, ready for visualization and DAX calculation.

6. Data Modeling

The project uses a dimensional modeling approach to organize the data efficiently for analytical queries.

6.1 Model Structure

The model contains one Fact Table (ad_events) and multiple Dimension Tables. While it follows many Star Schema principles, the additional relationship between the Ads and Campaigns tables — along with the Calendar Table — makes this a Hybrid Schema Structure rather than a pure Star Schema.

6.2 Table Relationships

From Table	To Table	Relationship Type
campaigns	ads	One-to-Many
ads	ad_events	One-to-Many
users	ad_events	One-to-Many
Calendar Table	ad_events	One-to-Many

7. Calendar Table

A dedicated Calendar Table was created using DAX based on the date range in the ad_events table. It unlocks Power BI's built-in Time Intelligence capabilities.

7.1 Calendar Table DAX

```
Calendar table =  
CALENDAR(  
    MIN(ad_events[event_date]),  
    MAX(ad_events[event_date])  
)
```

7.2 Calendar Columns

Column Name	DAX Formula	Description
Month	FORMAT([Date], "MMM")	Short month name (Jan, Feb, ...)
Day Name	FORMAT([Date], "ddd")	Abbreviated day name (Mon, Tue, ...)
Day Number	FORMAT([Date], "d")	Numeric day of the month
Week Day	WEEKDAY([Date], 2)	Day of week number (Mon=1)
Week Number	WEEKNUM([Date], 2)	ISO week number of the year

8. Calculated Columns

Two calculated columns were added to the ad_events fact table to enable time-based analysis.

8.1 Event Date

Extracts the date portion from the timestamp column for use with the Calendar Table.

```
event_date = DATEVALUE(ad_events[timestamp])
```

8.2 Event Hour

Extracts the hour from the timestamp column to enable hourly trend analysis.

```
Event Hour = HOUR(ad_events[timestamp])
```

9. Dynamic Measure Parameter

One of the most powerful features of this dashboard is the Dynamic Measure Parameter. A parameter table named 'Select Dynamic Measure' was created from the event_type column values from the ad_event table.

9.1 Available Event Types

The slicer allows users to dynamically switch the entire dashboard context to any of the following event types:

- Impressions
- Engagements
- Clicks
- Shares
- Comments
- Purchases

9.2 How It Works

When a user selects an Event Type from the slicer, the following elements update automatically across the entire dashboard:

- KPI Card headers change to reflect the selected metric
- Visual titles update dynamically via DAX measures
- Charts display data corresponding to the selected event type
- The entire dashboard context shifts to provide a focused analysis

Example: If the user selects 'Engagements', all KPI titles, chart headers, and visual insights will switch to engagement-focused metrics, creating a fully interactive analytical experience.

10. DAX Measures

10.1 Basic KPI Measures

These measures count specific event types from the `ad_events` table.

Measure	DAX Formula
Impressions	<code>COUNTROWS (FILTER (ad_events, ad_events[event_type] = "Impression"))</code>
Clicks	<code>COUNTROWS (FILTER (ad_events, ad_events[event_type] = "Click"))</code>
Comments	<code>COUNTROWS (FILTER (ad_events, ad_events[event_type] = "Comment"))</code>
Purchases	<code>COUNTROWS (FILTER (ad_events, ad_events[event_type] = "Purchase"))</code>
Shares	<code>COUNTROWS (FILTER (ad_events, ad_events[event_type] = "Share"))</code>
Engagements	<code>[Clicks] + [Shares] + [Comments]</code>

10.2 Budget Measures

Measure	DAX Formula
Total Budget	<code>SUM (campaigns [total_budget])</code>
Avg. Budget Per Campaign	<code>AVERAGE (campaigns [total_budget])</code>

10.3 Rate Measures

These measures calculate key advertising performance ratios.

Measure	DAX Formula	What It Tells You
CTR — Click Through Rate	<code>DIVIDE ([Clicks], [Impressions], 0)</code>	% of impressions that resulted in a click
ER — Engagement Rate	<code>DIVIDE ([Engagements], [Impressions], 0)</code>	% of impressions that generated engagement
CR — Conversion Rate	<code>DIVIDE ([Purchases], [Clicks], 0)</code>	% of clicks that converted to a purchase
PR — Purchase Rate	<code>DIVIDE ([Purchases], [Impressions], 0)</code>	% of impressions that resulted in a purchase

11. Dynamic Title Measures

These DAX measures automatically generate chart and visual titles based on the selected event type from the dynamic slicer. This ensures all titles stay in sync with the dashboard context.

Title Measure	DAX Formula
Age Title	SELECTEDVALUE('Select Dynamic Measure'[Dynamic Title]) & " by Age"
Gender Title	SELECTEDVALUE('Select Dynamic Measure'[Dynamic Title]) & " by Gender"
Hourly Title	"Hourly " & SELECTEDVALUE('Select Dynamic Measure'[Dynamic Title]) & " Trend"
Weekly Title	"Weekly " & SELECTEDVALUE('Select Dynamic Measure'[Dynamic Title]) & " Trend"
Map Title	SELECTEDVALUE('Select Dynamic Measure'[Dynamic Title]) & " by Country"
Matrix Title	"Analysis by Ad Type"
Calendar Title	"Analysis by Month"

12. KPI Cards

The dashboard features a comprehensive set of KPI Cards that display the most important advertising metrics at a glance.

KPI Card	Description
Impressions	Total number of times ads were displayed
Clicks	Total number of ad clicks recorded
Shares	Total number of ad shares
Comments	Total number of ad comments
Purchases	Total number of purchase conversions
Engagements	Combined total of Clicks + Shares + Comments
CTR	Click Through Rate — Clicks / Impressions
ER	Engagement Rate — Engagements / Impressions
CR	Conversion Rate — Purchases / Clicks
PR	Purchase Rate — Purchases / Impressions
Total Budget	Sum of all campaign budgets
Avg. Budget Per Campaign	Average budget across all campaigns

13. Dashboard Features

13.1 Platform Toggle Buttons

The dashboard includes two interactive toggle buttons — Facebook and Instagram — that allow users to filter all visuals and KPIs to a specific platform instantly. This enables quick side-by-side platform comparison and platform-level campaign analysis.

13.2 Tooltip KPI Feature

A custom tooltip page was created for the 'Analysis by Month' calendar visual. When a user hovers over a monthly data point, additional KPI details appear automatically, providing deeper comparative insights without cluttering the main dashboard.

13.3 Slicers

Three interactive slicers allow users to filter and refine the dashboard view:

Slicer	Filtering Options
Select Event Type	Impressions, Engagements, Clicks, Shares, Comments, Purchases
Campaign	Filter by specific campaign name
Target Interest	Art, Fashion, Finance, Food, Gaming, Technology, Photography, Sports, Travel

14. Visualizations

The dashboard uses a carefully selected set of visualizations to surface insights clearly and interactively.

Visual Type	Purpose & Details
KPI Cards	Display key advertising metrics at a glance
Donut Chart	Shows event distribution broken down by gender
Age Distribution Chart	Analyzes event performance across different age groups
Weekly Trend Chart	Displays how event performance varies across days of the week
Hourly Trend Chart	Analyzes event activity patterns across hours of the day
Map Visual	Displays country-wise event distribution geographically
Monthly Calendar Visual	Shows month-wise analysis in a calendar layout with tooltip KPIs
Matrix Table	Breaks down ad performance by Ad Type: Impressions, Clicks, CTR, ER, CR, PR

15. Business Insights

This dashboard empowers marketing teams and business stakeholders to make confident, data-driven decisions. Key use cases include:

- Analyze the overall effectiveness of ad campaigns across Meta platforms
- Compare Facebook and Instagram performance side by side
- Identify high-performing audience segments by age, gender, and interests
- Track engagement and conversion trends over time
- Optimize advertising budgets by identifying underperforming campaigns
- Deep-dive into platform-specific campaign performance
- Improve audience targeting strategies based on demographic insights
- Generate comprehensive, data-driven marketing reports

16. Key Learnings

This project strengthened practical skills across the full BI development lifecycle:

- Power BI Dashboard Development — end-to-end report design and publishing
- Dynamic Parameters — building flexible, user-driven dashboards
- DAX Calculations — writing custom measures for KPIs and rate metrics
- Data Modeling — designing efficient star/hybrid schema relationships
- Time Intelligence — applying CALENDAR tables and date functions
- KPI Analytics — defining, measuring, and visualizing business KPIs
- Interactive Reporting — tooltips, slicers, buttons, and dynamic titles
- Marketing Analytics — understanding advertising performance metrics

17. Conclusion

The Meta Ad Performance Analysis Dashboard successfully transforms raw advertising event data into meaningful marketing insights through interactive visualizations, dynamic KPI analysis, and advanced DAX calculations.

By combining a well-structured data model, a complete ETL pipeline, and dynamic reporting capabilities, the dashboard provides marketers and businesses with a powerful tool to:

- Understand how their ads are performing across Facebook and Instagram
- Identify which audiences, time periods, and ad types drive the best results
- Monitor key metrics — from impressions through to purchases — in a single view
- Make faster, more confident decisions backed by real advertising data

This project demonstrates a complete, production-ready Power BI solution that bridges the gap between raw data and strategic business decisions.